

M004 Week 4 — English Worksheet

Topic: Player Events · **Course:** Functions & Games · **Time:** about 45 minutes

This week your code reacts to **you**, the player. An **event** like `on player walk:` runs by itself when the player does something — no chat command needed. Perfect for traps and rewards.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think will happen.

```
on player walk:
  if block below is gold:
    say "Treasure room!"
```

The player walks across stone, stone, gold, stone. When does the message appear? How many times?


```
on player place block:
  say "You built something!"
```

The player places 3 blocks, then walks around for a while. How many times does the message appear? Does walking trigger it?


```
def drop_reward():  
    place diamond block  
  
on player walk:  
    if block below is redstone:  
        drop_reward()
```

The player steps onto a redstone block. What happens, step by step? What happens on normal ground?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — A trap message should appear **by itself** whenever the player walks onto gold. But nothing ever happens unless someone types `trap` in chat.

```
on chat command "trap":  
    if block below is gold:  
        say "Trap!"
```

Hint: this should be an **event** that runs on its own, not a chat command.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The player should get **one** welcome reward when they first step onto the redstone start tile. But the reward keeps firing again and again on every step, flooding the world with diamonds.

```
on player walk:  
    if block below is redstone:  
        place diamond block
```

Hint: use a variable like `reward_given` to remember if the reward already happened.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The message should appear when the player **places a block**, to celebrate building. But right now it appears every time the player simply walks.

```
on player walk:  
  say "Nice building!"
```

Hint: the behavior is right, the **event** is wrong.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a mini game zone with at least **one trap** and **one reward**, each triggered by a player event (`on player walk:` or `on player place block:`). When you finish, come back here.

Send a photo or video of your triggers firing, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My trap fires when the player ...

My reward fires when the player ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk through your game zone and set off the triggers. Talk through it like you are teaching someone who has never seen it. Try to use these words: **event**, **trigger**, **player**, **trap**, **reward**.

1. Show your game zone, then start walking.
2. Set off your **trap** and read its event block out loud.
3. Earn your **reward** and say which event made it fire.
4. Say in your own words how an **event** is different from a chat command.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.